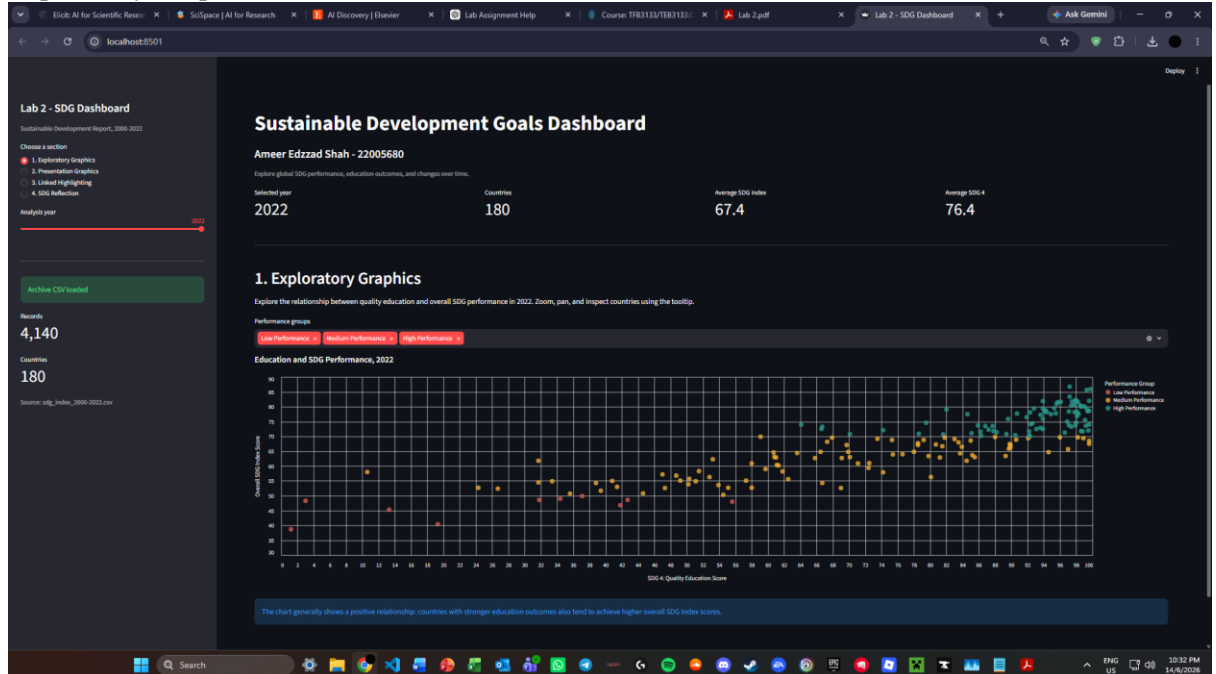


Name: Ameer Edzzad Shah

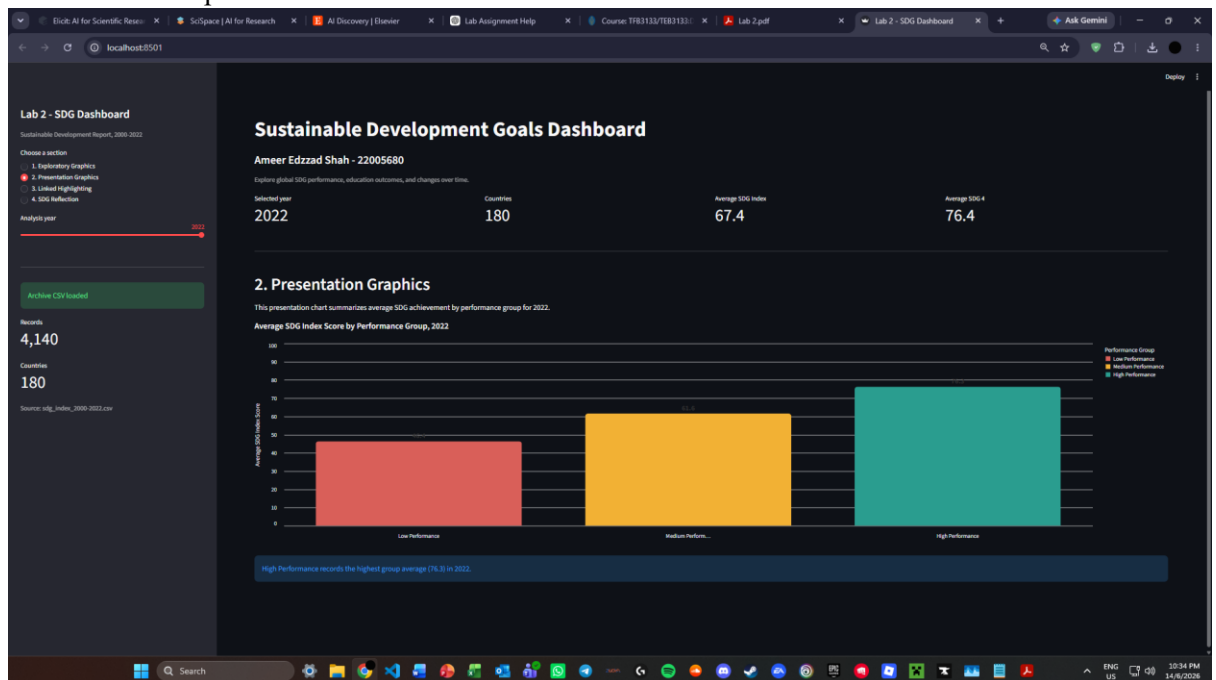
Student ID: 22005680

Group: G8

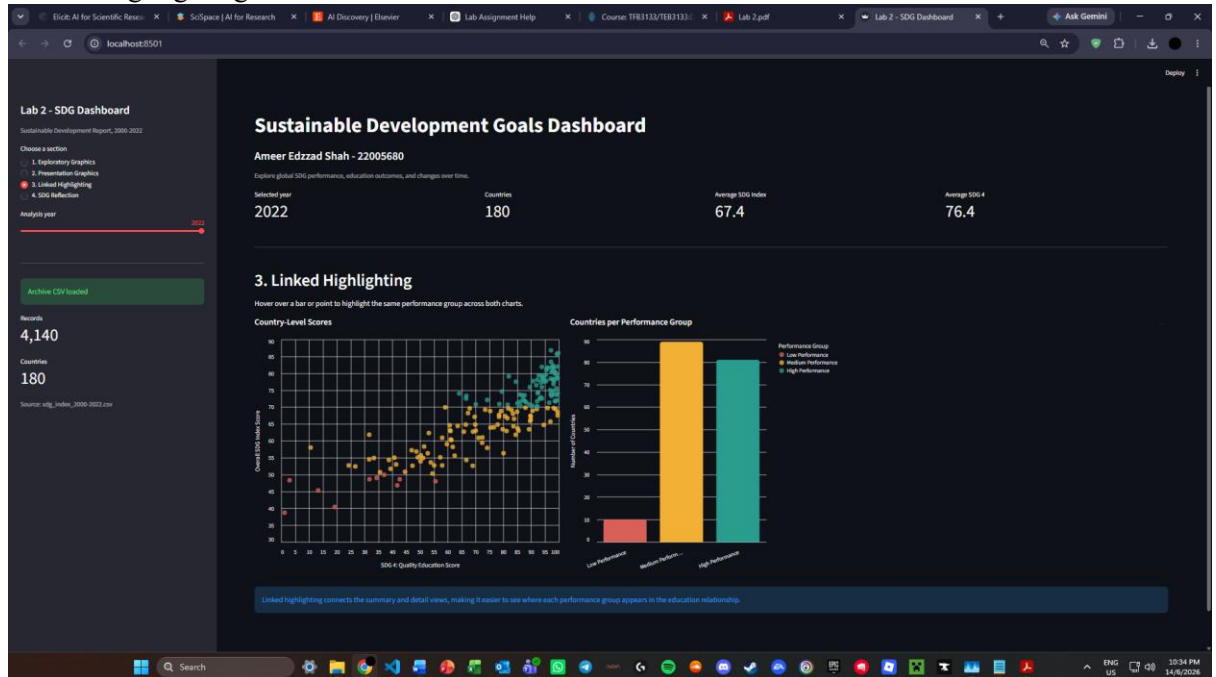
1. Exploratory Graphics:



2. Presentation Graphics:



3. Linked Highlighting:



4. SDG Reflection:



Short Report:

This dashboard was developed using Streamlit and Altair to explore the Sustainable Development Goals (SDG) dataset. Four visualizations were created to analyse relationships between countries, SDG performance, and education indicators.

The first visualization was an interactive scatter plot that compared SDG 4 (Quality Education) scores with overall SDG Index Scores. The chart showed a positive relationship between education performance and overall sustainable development. Countries with higher education scores generally achieved higher SDG Index Scores. The interactive tooltips allowed users to examine individual countries and their performance in greater detail.

The second visualization was a static bar chart showing the average SDG Index Score by performance group. The chart clearly demonstrated that countries classified under the high-performance group achieved significantly better SDG outcomes than countries in the medium and low-performance groups. This presentation graphic effectively communicated the overall differences in development performance.

The third visualization used linked highlighting to connect the scatter plot and bar chart. When a performance group was selected, the related countries were highlighted across both charts. This feature made it easier to identify patterns and relationships that would be difficult to observe using a single visualization. Linked highlighting improved data exploration by providing additional context and interaction.

The final visualization focused on SDG 4: Quality Education. A line chart was used to show changes in education scores over time. The results indicated that countries with stronger SDG performance generally maintained higher education scores throughout the observed period.

This highlights the importance of education as a foundation for sustainable development.

Quality education contributes to poverty reduction, economic growth, social equality, and improved quality of life.

Overall, the dashboard demonstrates how data visualization can be used to analyse sustainable development indicators and communicate insights effectively. The findings show that education plays a critical role in achieving broader SDG objectives and supporting long-term development across countries.